

Harnessing Smartphone Videos for Ground Reaction Force Estimation in Walking: A Deep Learning Approach

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INTRODUCTION

Ground reaction force (GRF) generated during gait cycle has shown differences between individuals with and without knee pain, which suggests their potential relationship in understanding knee osteoarthritis (KOA) symptoms[1]. Traditional GRF measurement techniques like force plates, are costly and usually limited to laboratory environments. In response to these challenges, we developed a smartphone-based approach that uses deep learning algorithms to estimate GRF from walking 2D videos.

METHODS

The gait videos were collected from 23 healthy participants (11 males, 12 females; mean age 58.04 years, range 19-72) walking at a preferred pace in our motion lab. 146 lateral-view walking videos were recorded using a smartphone camera from the right side, with each subject completing multiple trials. GRF data was collected through force plates serving as ground truth for the models, while a Vicon motion capture system recorded marker positions.

The dataset was divided by subjects, with 20 subjects used for training and 3 for testing. Among the training set, 48 videos utilized GRF data predicted from Vicon markers. 3D coordinates of 12 key joints were extracted from the videos using BlazePose, including shoulder, elbow, wrist, hip, knee, ankle, and foot markers. These joint trajectories were input into three deep learning architectures Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and Temporal Convolutional Network (TCN), and three hybrid models (CNN+LSTM, CNN+TCN, LSTM+TCN) to predict GRF values, with model performance assessed through RMSE and coefficient of determination.

RESULTS AND DISCUSSION

The hybrid CNN+LSTM model achieved the best performance ($R^2 = 0.86 \pm 0.08$), followed by the CNN ($R^2 = 0.82 \pm 0.09$) and LSTM model ($R^2 = 0.81 \pm 0.10$). The LSTM+TCN model showed moderate

performance ($R^2 = 0.75 \pm 0.08$), while the CNN+TCN ($R^2 = 0.70 \pm 0.08$) and TCN ($R^2 = 0.67 \pm 0.09$) models achieved lower performance.

Figure 1 shows the model effectively capturing the double-peak pattern in each gait cycle. The high R^2 values (>0.90) for both feet indicate strong correlation between predicted and measured forces. Despite small discrepancies in predictions, the overall performance demonstrates the potential of this smartphone-based method. Future work will focus on expanding the dataset to improve the model robustness.

Table 1. Performance comparison of different deep learning models for GRF prediction..

Model	R^2	RMSE (%BW)
CNN	0.82 ± 0.09	17.78 ± 4.50
LSTM	0.81 ± 0.10	18.34 ± 4.77
TCN	0.67 ± 0.09	24.57 ± 3.17
CNN+LSTM	0.86 ± 0.08	15.43 ± 4.49
CNN+TCN	0.70 ± 0.08	23.58 ± 3.20
LSTM+TCN	0.75 ± 0.08	21.32 ± 3.68

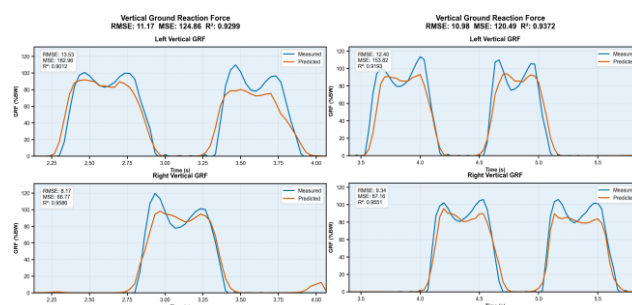


Fig. 1 – Comparison between measured and predicted vertical ground reaction forces (vGRF) during walking.

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